BETSYBOTS

DIY Challenge 2026

Team Kickoff — Development Strategy & Sprint Plan

ROS 2 Humble | FAST-LIO2 | Nav2 | LIO-SAM | Hesai QT64 | Orin Nano 8GB

June 2026

Speed & Focus

- Iterate on ROS2 nodes, Nav2 configs, launch files faster than from scratch
- Skip boilerplate: CMakeLists, package.xml, param YAML — let LLMs generate it
- Rapidly prototype ideas — cmd_vel logic, EKF tuning, costmap params

Where LLMs Excel

- EKF/Nav2 YAML params: LLMs reason about parameter interactions
- Targeted answers on FAST-LIO2 / LIO-SAM without reading all source
- Generate analysis scripts on demand — plotting, debugging, validation

Why This Matters for Us

- LLM-assisted development is day-to-day work at CAT — directly relevant practical training
- Prompting effectively for robotics code is a transferable engineering skill

Debugging With LLMs

Generate post-run analysis scripts on demand, iterate with LLM help

Trajectory Accuracy Plots

- Compare estimated path vs ground truth per lap
- Quantify drift per section — pinpoint where localization fails

EKF Covariance Health

- Plot innovation sequence over time
- Catch filter divergence before competition day

/cmd_vel Comparison

- /cmd_vel_safe vs /cmd_vel_nav overlaid in one time-series
- See if safety layer triggers unexpectedly mid-run

IMU Bias Drift

- Raw IMU vs EKF-corrected accel/gyro across a full run
- Data-driven justification to re-calibrate

Nav2 Costmap Animation

- Pull costmap snapshots from bag → animate → see if inflation radius or raytrace range causes path failures

Developer Work Split

One repo — task-based ownership, no folder locks

Developer 1 — Integration & Calibration

- System glue: startup sequence, cmd_vel mux, e-stop, URDF / robot description
- IMU calibration, LiDAR-IMU extrinsics, AprilTag EKF tuning on test track
- Repo management, branch health, integration testing, final stack validation

Developer 2 — Mapping & Localization

- Prior map generation via LIO-SAM offline mapping — joystick-driven on test track
- Validate FAST-LIO2 + RTK EKF fusion, measure lap-to-lap localization accuracy
- Slot calibration numbers into configs, verify improvement with fresh runs

Developer 3 — Perception & Mission Logic

- LiDAR cluster classifier: drive-through / go-around / ignore decisions
- Mission state machine: PUSH_THROUGH, NAVIGATE_AROUND, BLIND_DRIVE_MODE
- Offline bag replay so perception can be tested on a laptop without the robot

Other Focused Work Items

Smaller tasks — each needs an owner

Visual Start Detector	Camera node watching for race start signal → avoids 5-sec manual-start penalty
RViz2 Debug Config	Currently launches blank. Needs robot model, LiDAR scan, costmap, Nav2 path, TF tree
Competition Waypoints	Obstacle zone entries, narrow passage point, start/finish. Defined after prior map
Offline Bag Replay	Test perception on laptop without robot. Critical for Dev 3 between robot sessions
AprilTag EKF Tuning	Print tags, survey with RTK, drive laps, record bag, post-process to tune Q/R params

Why No Camera-Based Object Detection?

LiDAR-first perception is the right call for this outdoor challenge

Camera Problems Outdoors

- Sun glare, harsh shadows, exposure shifts
- Dust, reflections, lighting variance morning→afternoon
- Needs large labelled dataset — we don't have one
- Adds inference compute on already-busy Orin

LiDAR Geometry Wins

- Stable point cloud regardless of lighting
- Already running for odometry — zero extra cost
- PCL clustering is deterministic, no training data
- Map-diff catches new obstacles without any model

Camera Still Has a Role — Just Not Primary Perception

- AprilTag detection for EKF calibration on test track
- Visual start signal detector (avoid 5-sec penalty)
- Depth assist for chimes obstacle (LiDAR ground-level blind spot)

How We Work Together

Task-based ownership — small focused PRs, regular syncs

1

Pick a task → create a branch

Naming: feature/task-name · calib/imu-noise · fix/ekf-params

2

Own it end-to-end

No folder-level permission gates. Touch config, code, launch files — whatever the task needs.

3

Open PR → 1 review → merge

Small PRs scoped to one outcome. Describe changes, testing, and known risks.

4

Sync every 2 days (15 min max)

What changed · What outputs are ready to hand off · What blocks the next session

Branch Protection on main

- Direct pushes blocked — every change goes through a PR with ≥1 approval
- main is always launchable — nobody breaks the build for others

Week 1 — Calibration Lock

Make sensor geometry and noise trustworthy before anything else

Map quality, localization accuracy, and obstacle detection all depend on correct calibration. A bad calibration bakes errors into everything downstream — including the map itself.

CALIBRATION ORDER — must follow this sequence exactly

1 IMU Noise Calibration → **2 LiDAR-IMU Extrinsics** → **3 Camera-LiDAR Extrinsics** → **4 Record Prior Map**

✓ Definition of Done

- ✓ IMU noise values measured → applied to FAST-LIO config → committed
- ✓ LiDAR-IMU extrinsic transform → applied to config and URDF → committed
- ✓ Camera-LiDAR extrinsic transform → applied to URDF → committed
- ✓ Validation run: stable localization, no TF mismatch, no drift spikes
- ✓ All changes in one PR, reviewed and merged to main

Week 1 — Who Does What

Dev 1 — Integration Lead

- Prepare calibration branch & team checklist
- Verify all scripts run on robot + laptop before sessions
- Apply calibration outputs into config / URDF as numbers arrive
- Run final validation: TF tree, topic rates, no frame conflicts
- Open PR with test notes → review → merge
- Write tune_ekf_apriltag.py skeleton
- Build working RViz2 debug config

Dev 2 — IMU + LiDAR-IMU

- Run IMU noise calibration, extract noise params
- Update FAST-LIO noise values from results
- Run LiDAR-IMU extrinsic calib (figure-8 protocol)
- Deliver transform values to Dev 1
- Re-run to confirm FAST-LIO stable with new values

Dev 3 — Camera-LiDAR + Data QA

- Prepare calibration target board
- Run camera-lidar data collection session
- Process bag offline → solve transform
- Deliver transform values to Dev 1
- Record clean post-calibration validation bag

Jun → Aug 15 Roadmap

Competition Oct 1 — features locked by Aug 15 (45-day buffer for fixes & runs)

JUNE

Jun 1–28

Calibration & Mapping

- IMU noise + LiDAR-IMU extrinsics
- Camera-LiDAR extrinsics
- Prior map recording (LIO-SAM)
- FAST-LIO + RTK EKF validation
- AprilTag EKF tuning sessions

JULY

Jul 1–26

Perception & Nav Tuning

- LiDAR cluster classifier
- Zone-based behavior switching
- Nav2 param tuning (costmap, DWB)
- Offline bag replay workflow
- Lap accuracy < 10 cm drift

AUGUST

Aug 1–15

Mission Logic & Integration

- Mission state machine (zones)
- Visual start detector
- Competition waypoint set
- Full startup → run → stop flow
- E-stop stress testing

AUG 16 – SEP 30

45-day buffer

Bug Fixes & Practice Runs

- Multiple full-course dry runs
- Race-day runbook written
- Backup recovery procedures
- Hardware spares checked
- Nothing new — only fixes

Weekly Regroups — every Friday

- What passed consistently · what failed repeatedly · top 3 blockers for next week
- Aug 15 hard lock: no new features after this date — 45 days of bug fixes and practice runs only
- Oct 1: Competition day